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[Personal Research Website](#) [ResearchGate](#) [Google Scholar](#)

Research Interests

Ecohydrology; Biogeochemical Carbon Cycling; Ecosystem Resilience

Nature-based Climate Solutions; Natural Capital Valuation

Skills & Expertise

Data Analysis: R, MATLAB, Spatial Analysis, GIS

Optimization & AI: Machine Learning, Multi-objective Optimization

Modeling & Simulation: MIKE 11, Delft3D, ArcGIS, Hydrodynamic & Water Quality Models

Experimental & Analytical: Laboratory Rainfall Simulation, Spectroscopy, High-throughput Sequencing

Academic Performance

M.Eng. in Hydrology and Water Resources

Beijing Normal University, Beijing, China

Average score: 90.32/100

Rank: 1st among 96 students

Sept. 2021–Jun. 2024

Master's Thesis: Research on the characteristics of greenhouse gas emissions in the upper Yellow River basin and their potential association with dissolved organic matter composition

Main courses:

Water Quality Model and Simulation: A+ (Score: 96/100)

Application and Development of Spatial Analysis Technology in Environment: A+ (Score: 96/100)

Science of Hydrology and Water Resources: A (Score: 94/100)

Environmental Forensics: A (Score: 94/100)

Comparison of Environmental Issues between China and America: A+ (Score: 95/100)

B.Sc. in Geographical Science

Hunan Normal University, Changsha, China

Average score: 89.48/100

Rank: 1st among 84 students

Sept. 2017–Jun. 2021

Bachelor's Thesis: Runoff and sediment yield processes of two typical herbs on red soil slope under simulated rainfall

Main courses:

Economic Geography: A (Score: 94/100)

Meteorology and Climatology: A+ (Score: 95/100)

Natural Resources Science: A+ (Score: 98/100)

Geology and Geomorphology: A+ (Score: 95/100)

Soil Geography: A+ (Score: 95/100)

Phytogeography: A (Score: 91/100)

Language

English (fluent), Chinese (native)

TOEFL iBT: 101/120 (Test Date: 11/08/2025)

TOEFL iBT MyBest™ Scores: 107/120

Publications & Conferences

Published papers

[6] **Yang, Z.**, W, J., Li, X., Li, C., Pu, Z., Hu, J., Wang, X., Liu, Q., Yi, Y. 2025. An integrated framework for water assimilative capacity allocation based on environmental fairness-efficiency tradeoffs with a modified optimization model in a river basin. *Environmental Modelling & Software*, 191(9), 106503. DOI: [10.1016/j.envsoft.2025.106503](https://doi.org/10.1016/j.envsoft.2025.106503)

[5] **Yang, Z.**, Li, C., Liu, Y., Duan, J., Zhang, L., Li, Z., Zhou, X., Li, Q., Ma, Y., Tian, L., 2023. Roles of the stolon and erect grass species in surface–subsurface flow generation and red soil loss. *Journal of Hydrology*, 617, 128827. DOI: [10.1016/j.jhydrol.2022.128827](https://doi.org/10.1016/j.jhydrol.2022.128827)

[4] **Yang, Z.**, Bu, J., Li, Z., Li, C., Yi, Y., Wang, X., Liu, Q., 2023. A novel index-based method associated

- with aquatic ecosystem for evaluating river longitudinal connectivity: A case study for cascade dams in the Yalong River, China. *Ecological Indicators*, 154, 110903. DOI: [10.1016/j.ecolind.2023.110903](https://doi.org/10.1016/j.ecolind.2023.110903)
- [3] **Yang, Z.**, Li, C., Chen, H., Shan, X., Chen, J., Zhang, J., Liu, S., Liu, Q., Wang, X., 2023. Source-oriented ecological and resistome risks associated with geochemical enrichment of heavy metals in river sediments. *Chemosphere*, 336, 139119. DOI: [10.1016/j.chemosphere.2023.139119](https://doi.org/10.1016/j.chemosphere.2023.139119)
- [2] **Yang, Z.**, Han, L., Liu, Q., Li, C., Pan, Z., Xu, K., 2022. Spatial and temporal changes in wetland in Dongting Lake Basin of China under long time series from 1990 to 2020. *Sustainability*, 14(6), 3620. DOI: [10.3390/su14063620](https://doi.org/10.3390/su14063620)
- [1] Liu, Y., Ma, Y., Li, T., Xiao, P., Long, Y., Lei, M., **Yang, Z.**, Duan, J., Nie, X., Li, Z., 2025. Regulation mechanism of stolon and erect grass on surface–subsurface flow coupled erosion. *Journal of Hydrology*. 650, 132499. DOI: [10.1016/j.jhydrol.2024.132499](https://doi.org/10.1016/j.jhydrol.2024.132499)

Papers in prep. or under review

- [3] **Yang, Z.**, Li, C., Zhou X., Liu, L., Liu, L., Wang, X., Yi, Y., Liu, Q., 2025. A review of the global water resources under climate and land use changes in the past thirty years of research. *Hydrology and Water Resources*, Accepted (Manuscript ID: HWR-25080359).
- [2] Qu, Y., **Yang, Z.**, Gao, J., Liu, S., Li, C., Yi, Y., Wang, X., Liu, Q., 2025. Cascade reservoir development exerts greenhouse gas emissions through nutrient retention and modification of hydrological regimes. *Earth's Future*, Under Review (Manuscript Number: 2025EF007792).
- [1] Hu, J., **Yang, Z.**, Gao, J., Liu, M., Liu, S., Li, C., Yi, Y., Wang, X., Liu, Q., 2025. Enhanced Regulation of Carbon Emissions and DOM Transport by Cascade Reservoirs in the Upper Yellow River. *Journal of Hydrology*, Under Review (Manuscript Number: HYDROL75937).

Conferences

- The 5th International Conference of Geobiology, Wuhan, China, June 2025.
- The 2nd National Conference on Environmental Geoscience, Wuhan, China, April 2025.
- Poster Presentation, The 18th National Doctoral Academic Conference on Environment, Beijing, China, May 2024.
- Oral Presentation, The 18th National Graduate Forum on Environment, Shanghai, China, May 2023.
- Oral Presentation, The 1st Conference on World Geography, Shanghai, China, November 2022.
- Oral Presentation, The 6th River and Lake Ecological Forum, Beijing, China, May 2022.

Research Experiences

1. National Project: The Open Fund of the State Key Laboratory of Remote Sensing Science–Dynamic Response and Optimal Allocation of Water Environmental Capacity in the Luan River Basin, Chengde (No. 12800-310430011), Project Leader, May 2022–Dec 2023.

- Led research on basin-scale water environmental capacity response and optimal allocation under multiple regulation scenarios.
- Constructed and calibrated MIKE11 hydrodynamic–water quality models to simulate pollutant dynamics (COD, NH₃-N, TN, TP).
- Developed a fairness-oriented allocation framework using a modified environmental Gini coefficient and machine learning optimization, producing policy-relevant strategies for equitable water quality management.

2. Summer School Course: Frontiers of Geography–Carbon Neutrality and Climate Change, Peking University, 2023.

- Explored frontier research on carbon neutrality, including negative emissions technologies and carbon trading mechanisms.
- Applied GIS, remote sensing, and big data analytics to assess climate mitigation strategies and connect

scientific evidence with policy debates on resilience and global environmental governance.

3. Summer School Course: Theory and Methods of Land Surface Remote Sensing Inversion, Beijing Normal University, 2022.

- Studied land surface remote sensing inversion frameworks, including radiative transfer models and advanced retrieval methods for climate and ecosystem monitoring.
- Applied spectral analysis and satellite imagery interpretation to estimate soil moisture, vegetation indices, and energy fluxes, with frontier applications in carbon cycle and land–atmosphere interactions.

4. Provincial Research Project: The Spatiotemporal Change Characteristics and Driving Mechanisms of Wetlands in the Dong Ting Lake Basin, May 2019–May 2021.

- Investigated 30-year wetland dynamics in the Dongting Lake Basin using Landsat imagery and spatial analysis.
- Quantified land-use transitions and identified forestation and agricultural conversion as key drivers of wetland change under climate and human pressures.

5. Provincial Research Project: Location Modeling of Cultural and Creative Industries with Empirical Study in Hunan, Hunan Normal University, Jun 2018–Jun 2020

- Conducted GIS-based spatial analysis and land-use data interpretation to support location modeling of creative industry clusters.
- Contributed to empirical research on regional development patterns, integrating cultural geography and urban planning perspectives.

Honors & Awards

- Specifically reported on the official website of Beijing Normal University as a Distinguished Graduate ([Campus Featured Profile](#)).
- Baosteel Scholarship, Individual Award, 2023 (Top 0.002% nationwide, 500 annual awards for all graduate and undergraduate students in China).
- National Scholarship, Individual Award, 2023 (12,000 annual recipients in China).
- First-Class Scholarship, Beijing Normal University, Individual Award, 2023, 2022 (Rank No.1 in the Department of Environment Science).
- Bronze Award, the 5th China Universities Geography Science Exhibition Competition, South China, Group Award (Team Leader), 2021.
- Outstanding Graduate of Hunan Province, Individual Award, 2021.
- National Top 100 Practice Teams Award, the 5th Search for National College Students' Summer Practice Teams, Group Award (Team Leader), 2019.
- Second Prize, the 10th National Olympic English Essay Contest, Individual Award, 2015.
- First Prize, the 9th National "Earth Little Doctor" Geographic Science Popularization Knowledge Contest, Individual Award, 2015.

Service & Extracurricular Activities

- Conducted a 9-day summer field study in Tiaoma Village, Changsha, focusing on rural development, educational outreach, and local livelihood challenges as part of a university social practice team, 2018.
- Participated in a summer field study along the historic Tea-Horse Road in Anhua County, Hunan, examining rural livelihoods, cultural heritage, and development challenges, 2019.
- Participated in 38 volunteer service activities during the 2018–2019 academic year, accumulating 76 hours of community engagement.